Generalized Morse theory for distance functions

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1 Generalities on Lipschitz Functions

1.1 Lipschitz functions on manifolds and subdifferentials

1.1.1 General facts and definitions

For all the following, M will denote a differentiable manifold of dimension m. In this section, we generalize many definitions related to Lipschitz functions. We begin by extending the definition of Lipschitz functions to functions on a manifold.

Definition 1. *A function $f : M \to \mathbb{R}^n$, where M is a smooth manifold, is locally Lipschitz if for any chart $U \subset M$ and any compact $K \subset U$, there exists $C > 0$ such that, with the coordinates on U*

$$\forall x, y \in K, |f(x) - f(y)| \leq C|x - y|$$

Let $f : U \subset \mathbb{R}^m \to \mathbb{R}^n$ be a Lipschitz function, where U is open. Rademacher's theorem, which states that f is differentiable almost everywhere, will allow us to generalize differentials to this setting.

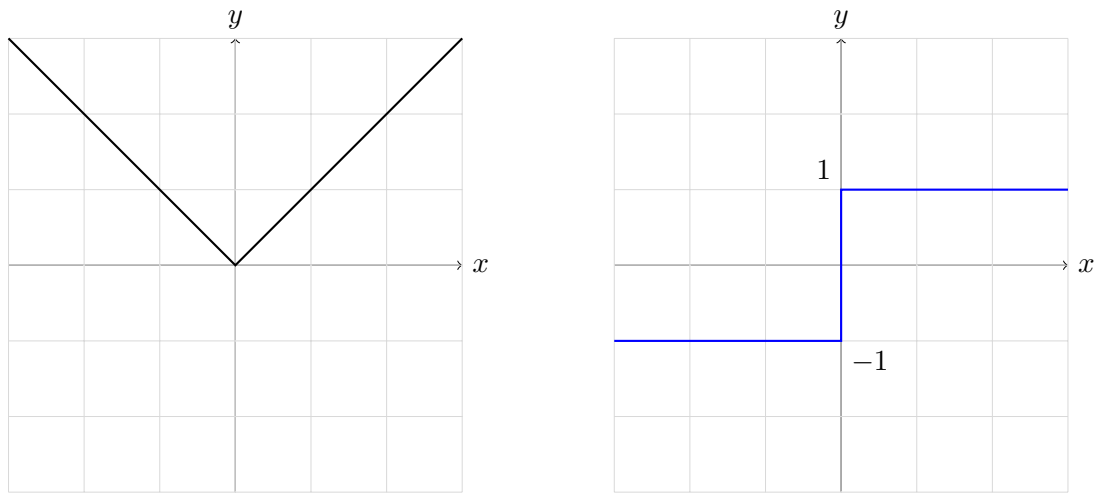
Definition 2. *[6] The subdifferential of f at x, denoted by $\partial_x f$ is the convex hull of the linear maps l obtained as*

$$l = \lim_{i \to \infty} d_{x_i} f$$

where f is differentiable at each x_i and $x_i \to x$.

The subdifferential is then set valued. This set is always non empty and compact, f being Lipschitz bounds df where it is defined.
By [5], we can exclude a set of measure 0 from the possible values of x_i without changing the definition.

Figure 1: The absolute value and its subdifferential represented in the plane
The set on the right is $\bigcup_{l \in \partial f} l$



Proposition 1. *∂f is the smallest convex valued upper semi-continuous function containing df whenever it is defined.*

Proof. First, ∂f is upper semi-continuous : by compactness, is suffices to show that for all $x \in U, \varepsilon > 0$, for y sufficiently close to x.

$$\partial_y f \subset \partial_x f + B(0, \varepsilon)$$

Assume for contradiction that there exists a sequence $x_i \to x$ such that for all i, $\partial_{x_i} f$ has an element out of $A := \partial_x f + \overline{B}(0, \varepsilon/2)$. A being convex and closed, let x_i' be such that $d_{x_i'} f$ is out of A, and such that $x_i' - x_i \to 0$. Up to extraction, $d_{x_i'} f$ converges to an element in $\partial_x f$, so it enters A, contradiction.
The second part follow from ∂f being a convex hull. $\square$

The subdifferential allows us to generalize the idea of regular and critical points, which will be an important tool for generalizing Morse theory. The definition is as follows :

Definition 3. *A point $x \in U$ will be called critical if there exists some $l \in \partial_x f$ not of maximal rank. x will be called regular if it isn't critical.*

To extend these subdifferentiability notions to manifolds, one must guarantee that for all charts, the definitions of ∂f coincide. In this is the case, $\partial_x f$ is a compact convex subset of $T_x^* M$. The following generalization of the chain rule allows us to do that.

Proposition 2 (Subdifferential chain rule)**.** *Let $h : \mathbb{R}^n \to \mathbb{R}^k$ be a C^1 function. For $x \in U$:*

$$\partial_x(h \circ f) \subset d_{f(x)}h \circ \partial_x f$$

If h is a diffeomorpism, we have equality.

Proof. For the first part, because f is differentiable a.e, it suffices to show that the leftwise term is an upper semi-continuous convex valued function that contains $d(h \circ f)$ whenever df is defined. This follows from the definition of ∂f.

If h is a diffeomorphism, then for $x \in U$:

$$\partial_x f = \partial_x(h^{-1} \circ h \circ f) \subset d_{h \circ f(x)}h^{-1} \circ \partial_x(h \circ f) \subset \partial_x f$$

so we have equalities, and the result follows. □

Although some important differentiablity related results can be obtained with locally Lipschitz functions as we'll see, a general locally Lipschitz function can have some pathological behaviours. As an example, it's subdifferential isn't in general a topological submanifold of M : define

$$A = \bigcup_{n \geq 0} \left[\frac{1}{2^{2n+1}}, \frac{1}{2^{2n}} \right[, B = \bigcup_{n \geq 0} \left[\frac{1}{2^{2n+2}}, \frac{1}{2^{2n+1}} \right[$$

The integral of the following function constitutes a counterexample, no chart can be found around the point $(0, 1)$ which is a point in the subdifferential.

$$\begin{array}{rcl} \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & \begin{cases} 2 & x \in A \\ 1 & x \in B \\ 0 & x \leq 0 \text{ or } x \geq 1 \end{cases} \end{array}$$

However, the functions we'll study, for example continuous selections (see Section 2) have much nicer behaviour and we retrieve this property.

1.1.2 The case of convex functions

If we consider convex Lipschitz functions $f : U \subset \mathbb{R}^m \to \mathbb{R}$, where U is a convex open set, the subdifferential can be understood much more precisely (in fact, this object was first invented for convex functions).

Consider a point $x \in U$ where $d_x f$ exists. By convexity, we have

$$\forall z \in U, f(z) \geq f(x) + d_x f(z - x)$$

Denote C_x the set of all linear maps verifying such property at a point x. The function $C : x \mapsto C_x$ is convex-valued, upper semi-continuous and contains d_x whenever it is defined. Moreover, at such points we have for h small enough :

$$f(x + h) = f(x) + d_x f(h) + o(h) \geq f(x) + \varphi(h)$$

where $\varphi \in C_x$. We deduce $(\varphi - d_x f)(h) = o(h)$, so $\varphi = d_x f$. Upper semi-continuity gives us $C \subset \partial f$, and the reciprocal is given by minimality of ∂f. So for a convex function the subdifferential can be described as

$$\partial_x f = \{\varphi \in (\mathbb{R}^m)^* \mid \forall z \in U, f(z) \geq f(x) + \varphi(z - x)\}$$

which gives a geometrical meaning to the name subdifferential.

Lastly, we connect the subdifferential to the directionnal derivatives of f. The directionnal derivative of f at x in direction $v \in \mathbb{R}^m$ is given by

$$D_x f(v) = \lim_{t \to 0^+} \frac{f(x + tv) - f(x)}{t}$$

and is always defined since f is convex. The two concepts are related by the following formula

$$D_x f(v) = \max_{l \in \partial f} \langle l, v \rangle$$

1.2 Generalized differential theorems

We now consider the specific case of $f : U \subset \mathbb{R}^m \to \mathbb{R}^m$. This section is devoted to proving generalizations by Clarke [6] of the inverse function theorem and the implicit function theorem for the case of Lipschitz functions.

In this section, we norm the space of linear maps $\mathbb{R}^m \to \mathbb{R}^m$, which we denote $\mathcal{L}(\mathbb{R}^n)$, with the operator norm $\|\cdot\|$ obtained from the norm used on $\mathbb{R}^n$.

Lemma 1. *Assume U is convex, and that there exists some closed convex set C such that*

- *Every $l \in C$ is invertible.*
- *$\partial_x f \subset C$ for all $x \in U$*

Then there exists some $\delta > 0$ such that for all $x, y \in U$:

$$|f(x) - f(y)| \geq \delta |x - y|$$

Proof. Firstly, we set

$$\delta = \inf \{|l(h)| \mid l \in C, h \in S\}$$

where S is the unit sphere in $\mathbb{R}^n$. $\delta > 0$ because all $l \in C$ are invertible and $C \times S$ is closed.

Let $x, y \in U$ and assume $x \neq y$. Let π be the plane perpendicular to $y - x$ passing through x. f being differentiable a.e, Fubini's theorem implies that for almost all $x' \in \pi$, f is differentiable almost on the whole line passing through x' and perpendicular to π (i.e f isn't differentiable on a subset of this line of one dimensional measure 0).
Denoting $F : t \longmapsto f(x' + t(y - x))$, we have :

$$f(x' + y - x) - f(x') = \int_0^1 F'(t)dt = \int_0^1 d_{x'+t(y-x)} f(y - x)dt \tag{1}$$

Define the linear map

$$L = \int_0^1 d_{x'+t(y-x)} f dt$$

C being convex and closed, $L \in C$. Observe that the integral on the right of (1) is just $L(y - x)$. So

$$|f(x' + y - x) - f(x')| \geq \delta |y - x|$$

This inequality holds for almost all $x' \in \pi$, so by continuity of f, it holds for x and we get the desired result. □

Theorem 1 (Lipschitz inverse function theorem)**.** *If x_0 is a regular point of f, then there exists V, W, neighbourhoods of x_0 and $f(x_0)$ respectively, such that f induces a bijection $f : V \to W$ where f^{-1} is Lipschitz too.*

Proof. By compactness of $\partial_{x_0} f$, regularity of x_0, and the fact that the invertible linear maps form an open set, there exist some $\varepsilon > 0$ such that the set

$$C := \partial_{x_0} f + \overline{B}(0, \varepsilon)$$

is composed only of invertible maps. By upper semi-continuity of ∂f, there is some $r > 0$ such that for $y \in B(x_0, r)$, $\partial f(y) \subset C$. We then apply Lemma 1 with $U = B(x_0, r)$ and get a corresponding $\delta > 0$.

Set $V = B(x_0, r/2)$ and $W = f(V)$. We showed in particular that f is injective on $\overline{V}$, by compactness $f : \overline{V} \to f(\overline{V})$ induces a homeomorphism. W is then open. The inequality we got with δ implies that f^{-1} is $\frac{1}{\delta}$ Lipschitz. □

From this theorem, follows the implicit function theorem for Lipschitz functions

Theorem 2. *Let $F : U \to \mathbb{R}^n$, where $U \subset \mathbb{R}^m \times \mathbb{R}^n$ is open and F is Lipschitz. Let $(x_0, y_0) \in U$ such that $F(x_0, y_0) = 0$ and y_0 is a regular point of $y \mapsto F(x_0, y)$. Then there exist neighbourhoods $(x_0, y_0) \in W \subset U$, $x_0 \in V \subset R^m$ and a Lipschitz function $g : V \to \mathbb{R}^n$ such that*

$$(x, y) \in W \text{ and } F(x, y) = 0 \Leftrightarrow x \in V \text{ and } y = g(x)$$

The proof is almost the same as the usual implicit function theorem, where the continuity of the differential is replaced by the semi continuity of ∂F.

1.3 First Morse lemma for locally Lipschitz functions

The goal of this section is to use the generalized notion of critical points in order to generalize the first Morse lemma to the case of locally Lipschitz functions. Fix M an m-dimensional smooth manifold, $f : M \to \mathbb{R}$ a continuous function and define for all $c \in \mathbb{R}$ the sublevel set of f of value c as

$$M_c := f^{-1}(]-\infty, c])$$

The classical first Morse lemma states that if f is C^1, and there are no critical points in $f^{-1}([a, b])$ (that we assume to be compact), then M_b is diffeomorphic to M_a and M_a is a deformation retract of M_b. We extend this to locally Lipschitz functions, following [1]. Note that in this setting x is a critical point of f iff $0 \in \partial_x f \subset T_x^* M$.

Proposition 3. *Let $a < b \in \mathbb{R}$ such that $f^{-1}([a, b])$ is compact and free of critical points. Then there exists a locally Lipschitz homotopy $H : M_b \times [0, 1] \to M_b$ such that for all $s \in [0, 1]$, $H(\cdot, s)$ is a homeomorphism verifying*

$$H(M_b \times \{s\}) \subset M_{b(1-s)+as}$$

Proof. We'll construct $V \in \mathrm{Vec}(M)$ a smooth complete vector field on M such that, for all $x \in M$ such that $d_x f$ is defined :

$$d_x f(V(x)) \leq 0 \text{ and } d_x f(V(x)) \leq -1 \text{ if } x \in f^{-1}([a, b]) \tag{2}$$

Let $x \in f^{-1}([a, b])$. x being regular and $\partial_x f$ convex, we find an affine hyperplane in $T_x^* M$ separating it from 0. This gives us $v_x \in T_x M$ such that $\langle \partial_x f, v_x \rangle \leq -2$. By semi continuity, there exists an open neighbourhood U_x of x such that for all $y \in U_x$, $\langle \partial_y f, v_x \rangle \leq -1$. We choose this neighbourhood such that it has compact closure.

These U_x cover $f^{-1}([a, b])$. Let $U_{x_1}, ..., U_{x_r}$ be a finite subcover, and let $\rho_1, ..., \rho_r$ be a partition of unity subordinated to this covering. We set

$$\forall x \in M, V(x) = \sum_{i=1}^{r} \rho_i(x) v_{x_i}$$

which indeed verifies (2) : if $d_x f$ is defined, it is contained in $\partial_x f$, and V is of compact support so it is complete.

Now observe that for $x \in f^{-1}([a,b])$, $t \mapsto f(e^{tV}x)$ is strictly decreasing. Denote by $t(x,s)$, $s \in [0,1]$, the unique time such that $f(e^{t(x,s)V}x) = as + f(x)(1-s)$. The equation defining $t(x,s)$ is locally Lipschitz in s, x and t, so by the Lipschitz implicit function theorem, t is a locally Lipschitz function of (x,s).

We can similarly define t' a locally Lipschitz function of x such that $f(e^{t'(x)V}x) = b$ for x in the orbit of some $y \in f^{-1}(b)$ by V. Denote O the set of $x \in M$ that are in such orbits. The homotopy is then given by

$$H : \begin{array}{ccc} M_b \times [0,1] & \longrightarrow & M_b \\ (x,s) & \longmapsto & \begin{cases} e^{t(e^{t'(x)V}x,s)V}x & \text{if } x \in O \\ x & \text{else} \end{cases} \end{array}$$

$\square$

2 Morse theory of continuous selections

2.1 Definitions

We now restrict ourselves to continuous functions that are piecewise smooth with finitely many pieces. We define those as continuous selections.

Definition 4. *A continuous function $f : M \to \mathbb{R}$ is a continuous selection if there exists $\varphi_1, ..., \varphi_n$ "regular" functions such that*

$$\forall x \in M, f(x) \in \{\varphi_1(x), ..., \varphi_n(x)\}$$

The term "regular" is used loosely here to indicate that the φ_i satisfy some extra condition on top of being continuous. For what follows, we'll restrict ourselves to continuous selections of smooth functions (but C^2 will suffice).

Note that a continous selection of locally Lipschitz functions is itself locally Lipschitz : if M is an open set of $\mathbb{R}^m$, and we assume that the functions φ_i are Lipschitz with constants c_i, then f is Lipschitz with constant $\max_i c_i$. Because smooth functions are locally Lipschitz, the functions we'll study will be too.

We can then relate ∂f to the differentials of the φ_i. Indeed, a limit of points of the form $d_x\varphi_i$ in T^*M is of the same form. Define the set of active indices at x,

$$I(x) = \{i \in [\![1, n]\!] \mid \varphi_i(x) = f(x)\}$$

$\partial_x f$ is then the convex hull of some $d_x\varphi_i$, $i \in I(x)$. It is possible that not all $d_x\varphi_i$ are used, if for example one of the functions isn't active in a neighbourhood of x.

We would like to define what it means for such a continuous selection to be Morse. Let f be a continuous selection of $\varphi_1, ..., \varphi_n$. Following [1], we give three conditions for a function to be considered Morse.

$$\text{The graphs of the } \varphi_i, i \in [\![1, n]\!], \text{ are transversal} \tag{C1}$$

This condition has some important consequences : for a non empty $I \subset [\![1, n]\!]$, we define $W_I := \{x \in M \mid I(x) = I\}$. Condition (C1) implies that W_I is a submanifold of M of codimension $|I| - 1$: the intersection of the graphs of $\varphi_i, i \in I$ is a submanifold of $M \times \mathbb{R}$ of codimension $|I|$ and W_I is the pre-image of this intersection through the diffeomorphism $x \mapsto (x, \varphi_i(x))$, $i \in I$, with the codomain being the graph of φ_i.

This transversality condition also implies that for $x \in M$, the differentials $d_x\varphi_i$ are affinely independant where $i \in I(x)$. By affinely independant, we mean that one of the two equivalent conditions are satisfied : for some/any $j \in I(x)$, the family $(d_x\varphi_i - d_x\varphi_j)_{\substack{i \in I(x) \\ j \neq i}}$ is linearly independant, or equivalently, the family

$$\begin{pmatrix} d_x\varphi_i \\ 1 \end{pmatrix}, i \in I(x)$$

is linearly independant. This implies that $\partial_x f$ is a simplex with less than $|I(x)| - 1$ vertices.

Note that for all $I \subset [\![1, n]\!]$, $f_{|W_I}$ is a smooth function. This motivates the second condition

$$\text{For all non empty } I \subset [\![1, n]\!], f_{|W_I} \text{ is Morse} \tag{C2}$$

Consider $x \in M$ a critical point of f. Observe that x must also be a critical point of $f_{|W_{I(x)}}$. So $T_x W_{I(x)} \subset \bigcap_{i \in I(x)} d_x\varphi_i$, but the converse can fail if the $d_x\varphi_i$ are not linearly independant enough. The last condition fixes that

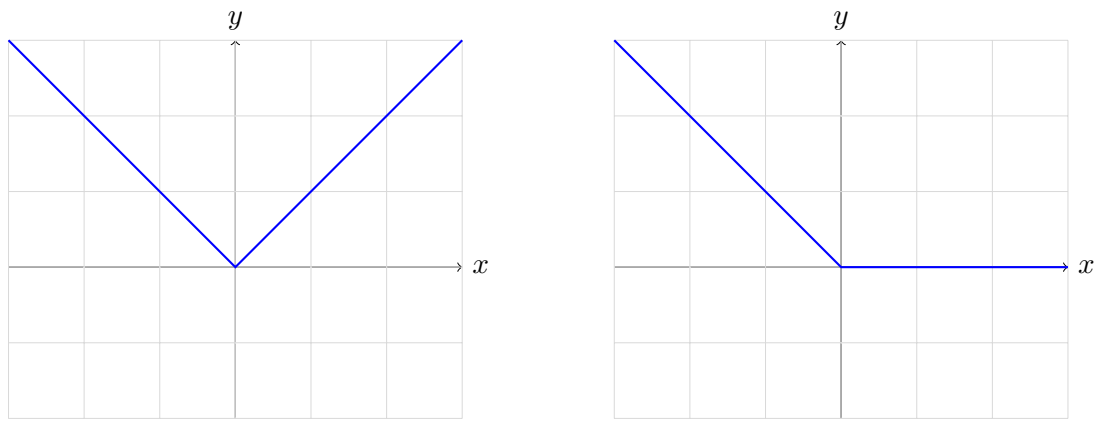
$$\forall x \in M, \forall j \in I(x), (d_x\varphi_i)_{\substack{i \in I(x) \\ i \neq j}} \text{ are linearly independant} \tag{C3}$$

Geometrically, this condition corresponds to requiring that 0 is either outside or in the interior of the simplex $\mathrm{conv}\{d_x\varphi_i \mid i \in I(x)\}$, but not on one of its faces. Note that this condition can be relaxed by restricting it to the subset of $I(x)$ participating in $\partial_x f$.

A continuous selection f is then called a Morse function if it verifies all three conditions. We can also impose that these conditions only hold at critical points, and we get the same theory.

Note that as in the smooth case, a generic continuous selection is Morse, i.e the set of all n-tuples $(\varphi_1, ..., \varphi_n)$ such that these conditions hold at every critical point is open dense in C^2 topology [7]. We can see it intuitively by noticing that (C1) is a generic condition, any two functions can be slightly perturbed in order for their graphs to be transversal, and any small perturbation of transversal graphs preserves transversality. By genericity of smooth Morse functions, (C2) is also a generic condition. With the geometric interpretation, we can also see that condition (C3) is generic.

Figure 2: Example and counterexample of condition (C3)



2.2 Normal form of a Morse continuous selection

We show here an equivalent of the second Morse lemma for the case of Morse continuous selections. Noticing that $x \mapsto I(x)$ is upper semi-continuous and using the definitions yields the following result.

Lemma 2. *If $f : M \to \mathbb{R}$ is a Morse continuous selection, all its critical points are isolated.*

We'll then consider, up to restricting ourselves to a small enough chart, that $f : \mathbb{R}^m \to \mathbb{R}$ is Morse and has no critical points other than 0. We'll write for shorthand $W := W_{I(0)}$, $k := |I(0)| - 1$, $c := f(0)$, and assume that $I(0) = [\![1, k+1]\!]$. Semi-continuity of the active index also allows us to assume that for $x \in \mathbb{R}^m$, $I(x) \subset [\![1, k+1]\!]$.

Notice that because of condition (C1) and semi-continuity of the active index, the family $(W_I)_{I \subset [\![1,k+1]\!]}$ gives a Whitney stratification of $\mathbb{R}^m$. This observation motivates the following lemma, in the spirit of Thom isotopy theory.

Lemma 3. *Let $h : \mathbb{R}^k \times \mathbb{R} \to \mathbb{R}$ be a Morse continuous selection. There exists a neighbourhood U of 0 and $\varepsilon > 0$, a vector field $X : U \times]-\varepsilon, \varepsilon[\to \mathbb{R}^k \times \mathbb{R}$ verifying*

1. *For $x \in U, t \in]-\varepsilon, \varepsilon[$ we have*
$$X(x,t) = (\tilde{X}(x,t), 1)$$

2. *If $(x,t) \in W_I$ for some active index set I, then X is tangent to W_I at all times and the restriction to W_I is C^1*

3. *$(x,t) \mapsto h(x,t)$ is constant along integral curves of X*

4. The flow of X is continuous.

The use of this lemma requires a time dependent continuous selection. For the proof of the main theorem, we will have to build one with some boundary conditions wich is done in the following way

Lemma 4. *Let $\varphi_1, ..., \varphi_{k+1} : \mathbb{R}^k \times \mathbb{R} \to \mathbb{R}$ be smooth functions verifying conditions* (C1) *to* (C3) *and $h^0 : \mathbb{R}^k \to \mathbb{R}$ be a continuous selection of $\varphi_1(\cdot, 0), ..., \varphi_n(\cdot, 0)$. Then there exist U a neighbourhood of* 0 *and a continuous selection $h : U \times \mathbb{R} \to \mathbb{R}$ of $\varphi_1, ..., \varphi_{k+1}$ such that $h(\cdot, 0) = h^0$.*

Proof. We denote by W_I the submanifold of $\mathbb{R}^k$ where the active index set of h^0 is exactly I. We restrict ourselves to an open neighbourhood of 0 in $\mathbb{R}^k$ of compact closure which we call U. Our goal will be to partition U in a specific way, and then trivially define h with this decomposition.

For all $i \in [\![1, k+1]\!]$ and $I \subset [\![1, k+1]\!]$ define

$$R_I^i = \{(x,t) \in U \times \mathbb{R} \mid \{j \mid \varphi_i(x,t) = \varphi_j(x,t)\} = I\}$$

Note that $W_I \subset R_I^i$ but that the slice $R_I^i \cap \mathbb{R}^k$ can be larger than W_I. We remedy that by noticing that each R_I^i is a submanifold of $U \times \mathbb{R}$ with no boundaries. We then find for all W_I a neighbourhood S_I of it in R_I^i where $i \in I$. This partitions $U \times J$, where J is an interval containing 0 obtained by compactness. It now suffices to define $h(x,t)$ as equal to $\varphi_i(x,t)$, where i is such that $i \in I$, $(x,t) \in S_I$. This ensures continuity of h and the fact that it coincides with h^0.

We now want to extend this selection on the whole real line. This is a classical connectedness argument : each φ_i is defined on the whole real line so if we consider an open neighbourhood of 0 where the selection is defined, then we can extend h to the boundary continuously. Hence h can be defined on the whole real line times U. □

We are now ready to prove the main result of this section.

Theorem 3 (Normal form)**.** *If* 0 *is a critical point of f, then there exists continuous local coordinates around* 0*, $\psi : \mathbb{R}^k \times \mathbb{R}^{m-k} \to U$ such that*

$$\forall y \in \mathbb{R}^k, \forall z \in \mathbb{R}^{m-k}, f(\psi(y,z)) = \lambda(y) + q(z)$$

where λ is a continuous selection of $k+1$ affinely independant linear functions, and q is of the form

$$\forall z \in \mathbb{R}^{m-k}, q(z) = c - \sum_{i=1}^{\mu} z_i^2 + \sum_{i=\mu+1}^{m-k} z_i^2$$

with $0 \leq \mu \leq m-k$.

Proof. We first restrict ourselves to O a neighbourhood of 0 such that W can be considered rectified in some smooth coordinates i.e, $W \cap O = 0 \times \mathbb{R}^{m-k}$. We can also use the smooth Morse lemma to get $f(0,z) = q(z)$. For $(y,z) \in \mathbb{R}^m$, write $f^0(y,z) = f(y,z) - f(0,z)$. We will build for all z close enough to 0 a locally invertible map $\psi_z : \mathbb{R}^k \to \mathbb{R}^k$ such that $\psi_z(0) = 0$ and for y close enough to 0, $f^0(\psi_z(y), z)$ is a continuous selection of linear maps independant of z.

For what follows we'll fix z and omit it in the notations for convenience. We also assume without loss of generality that $f(0) = 0$, up to replacing it by f^0.

First, by condition (C3), we can by the implicit function theorem assume that the k first φ_i are linearized, i.e equal to $d_0\varphi_i$ in a neighbourhood of 0. We want to linearize the last one and we will be done. For that, for all $t \in [0,1]$ define

$$\varphi_{k+1}^t = (1-t)\varphi_{k+1} + t d_0\varphi_{k+1}$$

By lemma 4, we define $h(y,t)$ for all y close enough to 0 and t as a continuous selection of $d_0\varphi_1, ..., d_0\varphi_k, \varphi_{k+1}^t$ such that it equals f at $t = 0$. This is Morse for y close enough to 0, as the

differential of φ_{k+1}^t for such y is close to the one of φ_{k+1}.

We then apply lemma 3 and get U an even smaller neighbourhood and X a vector field on a neighbourhood of $(0,0)$. Up to shrinking U, we can assume that for all t^* there is a time $s(t^*) > 0$ such that for all $y \in U$, we can integrate (y, t^*) for time $s, |s| < s(t^*)$. To give some notation, the result of this integration, $\Phi(s, y, t^*)$, is well defined. This gives a covering of $[0,1]$ by intervals of the form $]t^* - s(t^*), t^* + s(t^*)[$, so we can still cover it with finitely many of these intervals.

Assume for simplicity that we can flow for time 1 from all $(y, 0)$, $y \in U$ i.e $[0,1]$ can be covered with only one such interval. We construct a local homeomorphism taking $h(y,0)$ to $h(y,1)$. Set

$$\psi(y', t) = \Phi(-t, y', t) + (0, ..., 0, t)$$

We can then check by using properties 1. and 3. that $h \circ \psi^{-1}(y, 1) = h(y, 0)$. To sum up, $f \circ \psi(y, 0) = h(y, 1)$ i.e a continuous selection of $k+1$ affinely independant linear functions. If we need more than one interval to cover $[0,1]$ we can decompose this into finitely many homeomorphisms going from 0 to 1 in successive steps.

We constructed ψ_z for all z close enough to 0 to approximate the behaviours of the φ_i by what happens at $(0,0)$. We can also easily see that if we add back the z, the construction is continuous in z too. Then $(y, z) \mapsto (\psi_z(y), z)$ is a local homeomorphism, and we have what we want.

□

We now try to understand better the structure of the non smooth component that appears in the normal form theorem. It turns out that it can be encoded in a purely combinatorial way, as the following theorem shows [9].

Theorem 4. *Let $\lambda : \mathbb{R}^k \to \mathbb{R}$ be a continuous selection of $k+1$ affinely independant linear functions, $\lambda_1, ..., \lambda_{k+1}$. Then there exists a unique family $S_1, ..., S_l$ of subsets of $[\![1, k+1]\!]$ verifying that*

- *$S_i \not\subset S_j$, $i \neq j$*
- *λ can be written as*

$$\lambda(y) = \min_i \max_{j \in S_i} \lambda_j(y)$$

Proof. Let's first tackle uniqueness. Let $S_1, ..., S_l$, $D_1, ..., D_r$ be two such distinct families. Assume without loss of generality that $S_1 \neq D_j$. We can also assume that for all $1 \leq j \leq r$, $S_1 \not\supset D_j$. If it is the case, then for all $1 < i \leq l$ we have $D_j \not\supset S_i$ by assumption on S. We can then swap S and D to get what we want.
By affine independance, the following linear system has a solution

$$\lambda_i(y) + \mu = c_i, i \in [\![1, k+1]\!]$$

where the unknowns are y, μ and $c_i = 0$ if $i \in S_1$ and 1 otherwise. It follows that, if we use the representation with D_i, we get $\lambda(y) = 1 - \mu$ and if we use the one with the S_i, $\lambda(y) = -\mu$.

Now for existence. For all permutation $\nu \in \mathfrak{S}_{k+1}$ define the the cone

$$K_\nu = \{y \in \mathbb{R}^k \mid \lambda_{\nu(1)}(y) \leq ... \leq \lambda_{\nu(k+1)}(y)\}$$

and the function $s : \nu \in \mathfrak{S}_{k+1} \mapsto i \in [\![1, k+1]\!]$ where i is such that $\lambda_{\nu(i)} = \lambda$ on K_ν. If we assume that for all y and ν we have

$$\lambda(y) \leq \max\{\lambda_{\nu(1)}(y), ..., \lambda_{\nu(s(\nu))}(y)\}$$

then it suffices to consider the family of set $S_\nu = \{\nu(1), ..., \nu(s(\nu))\}$ and simplify these by removing those who are included in one another.

We now give a broad idea of the proof of the claim. The result is true by definition on K_ν. For any $y \in \mathbb{R}^k$, set $y' \in K_\nu$ and consider the evolution of the order of λ_i along the line between y and y' : the

value of each λ_i evolves at a constant rate. Whenever the current λ_i defining λ encounters another λ_j and λ becomes defined as λ_j afterwards, either λ_j is smaller than λ_i after the encounter, or the opposite. If we denote I the set of indices such that the functions are smaller than λ_j before the encounter, then in either case we see that λ stays smaller than the maximum of $\lambda_{i'}$ $i' \in I$. This concludes the proof because the line starts at K_ν. □

2.3 Describing the homology of sublevel sets

Our goal for this section is to describe more precisely the transition that happens as the critical point of f is passed, where f is still a Morse continuous selection defined on $\mathbb{R}^m$ with 0 as its only critical point.

The normal form of f and theorem 4 allows us to fully describe the behaviour of f with purely combinatorial information, a smooth Morse index and a more complicated set representation. We can encode the combinatorial structure that appears in its non smooth part in a way that will give us topological and homotopical informations about the sublevel sets : define the simplical complex $K(S_1^c, ..., S_l^c)$ on $k+1$ vertices whose faces are given by the complements in $[\![1, k+1]\!]$ of the sets S_i describing the non smooth part of f.

Another thing to note is that when put in normal form, f is a quasi homogeneous function, i.e there exists some real numbers $\alpha_1, ..., \alpha_m > 0$ such that for all $t > 0, x \in \mathbb{R}^m$, we have

$$f(t^{\alpha_1} x_1, ..., t^{\alpha_m} x_m) = tf(x)$$

In this case we have $f(ty, t^{1/2} z) = tf(y, z)$ if we assume that $f(0) = 0$, assumption we'll make without loss of generality. For a quasi homogenous function h, we'll denote δ_t the power-scaling map such that $h(\delta_t x) = th(x)$. The following lemmas on quasi homogeneous functions will prove useful for describing the homotopy type of sublevel sets.

Lemma 5. *Let $h : \mathbb{R}^m \to \mathbb{R}$ be a quasi homogeneous locally Lipschitz function having only 0 as a critical point. We have the following homotopy equivalence*

$$(M_1, M_{-1}) \simeq (C_{h^{-1}(-1)}, h^{-1}(-1))$$

where M_1, M_{-1} are the sublevel sets and $C_{h^{-1}(-1)}$ is the cone over $h^{-1}(-1)$.

Proof. Let $\mathbb{S}$ denote the unit sphere. Define a smooth V over $\mathbb{S}$ such that for all $x \in \mathbb{S}$, $V(x) \in \mathbb{R}^m$ and $\langle \partial_x h, V(x) \rangle \leq -1$. Extend V into a vector field on $\mathbb{R}$ such that it commutes with δ_t for all $t > 0$, i.e for $x \in \mathbb{S}$, $V(\delta_t x) = d_x \delta_t(V(x))$. It follows that $s \mapsto h(e^{sV} x)$ is strictly decreasing for all $x \neq 0$.

From this observation, we define similarly to proposition 3, for all $x \in h^{-1}(]-\infty, 0[)$, $p(x)$, the only point of the orbit of x in $h^{-1}(-1)$. We also define for all $x \in h^{-1}(-1), s \in [0, 1[$, $t(x, s)$ such that it verifies $h(e^{-t(x,s)V} x) = h(x) + s$. If we then denote by $\pi : [0, 1] \times h^{-1}(-1) \to C_{h^{-1}(-1)}$ the projection map, where the top of the cone is the point at height $s = 1$, the homotopy equivalence maps are the following

$$F : \begin{array}{ccc} M_1 & \longrightarrow & C_{h^{-1}(-1)} \\ x & \longmapsto & \begin{cases} \pi(0, p(x)) & x \in M_{-1} \\ \pi(h(x) + 1, p(x)) & x \in h^{-1}([-1, 0]) \\ \pi(1, 0) & x \in h^{-1}([0, 1]) \end{cases} \end{array}$$

$$G : \begin{array}{ccc} C_{h^{-1}(-1)} & \longrightarrow & M_1 \\ \pi(s, x) & \longmapsto & e^{-t(x,s)} x \end{array}$$

We can see that $F \circ G = \mathrm{id}_{C_{h^{-1}(-1)}}$. $G \circ F$ is homotopic to the identity : we can use V to contract M_{-1} onto $h^{-1}(-1)$, and also to retract $h^{-1}([0, 1])$ onto $h^{-1}(0)$. □

If X and Y are two topological spaces, we define the join of X and Y, denoted by $X * Y$, as the quotient space of $X \times Y \times [0,1]$ by the following equivalence relation

$$\forall x, x' \in X, \forall y, y' \in Y, (x, y, 0) \sim (x, y', 0) \text{ and } (x, y, 1) \sim (x', y, 1)$$

Intuitively, we join together the spaces X and Y with all possible segments between the two. Some examples of join include $X * \{y\}$ wich is the cone of X, $X * \mathbb{S}^0$ which is the suspension of X. More generally the join of X with $\mathbb{S}^n$ is an $n+1$-fold suspension.

By using the same ideas as in lemma 5, one can prove the following useful lemma.

Lemma 6. *Let $h : \mathbb{R}^k \times \mathbb{R}^{m-k} \to \mathbb{R}$ be a quasi homogeneous function such that for all $(y, z) \in \mathbb{R}^m$, we have $h(y, z) = a(y) + b(z)$ where a and b are quasi-homogeneous locally Lipschitz with respective exponents $\alpha, \beta > 0$. We have the following homotopy equivalence*

$$h^{-1}(-1) \simeq a^{-1}(-1) * b^{-1}(-1)$$

If we write $f(y, z) = \lambda(y) + q(z)$ as described by the normal form theorem, what remains to be done to understand the relative homology of (M_1, M_{-1}) is to describe $q^{-1}(-1)$ and $\lambda^{-1}(-1)$. The first one has the homotopy type of $\mathbb{S}^\mu$ where μ is the smooth Morse index of f, by classical Morse theory.

For the second, note that we have the following homotopy equivalence $\lambda^{-1}(-1) \simeq \lambda^{-1}(]-\infty, 0[)$. Using the min max form, we get

$$\lambda^{-1}(]-\infty, 0[) = \bigcup_i \bigcap_{j \in S_i} \lambda_j(]-\infty, 0[)$$

Note that each intersection is a convex cone we'll denote by C_i. We'll relate this to the simplical complex $K(S_1^c, ..., S_l^c)$ by using the nerve of the family $C = (C_i)_i$, i.e we'll look at the following set

$$N(C) := \{J \subset [\![1, l]\!] \mid \bigcap_{j \in J} C_j \neq \varnothing\}$$

From the next lemma, it follows that the nerve of $N(C)$ is exactly the one of the family of sets $S_1^c, ..., S_l^c$.

Lemma 7. *Let $J \subset [\![1, k+1]\!]$. J is the whole set if and only if*

$$\bigcap_{j \in J} \lambda_j^{-1}(]-\infty, 0[) = \varnothing$$

Proof. We first show that if $J = [\![1, k+1]\!]$, we indeed get an empty intersection. Recall that 0 is a critical point of f, so it is a critical point of λ. We then have

$$0 \in \partial_0 \lambda \subset \operatorname{conv}\{\lambda_1, ..., \lambda_{k+1}\}$$

So there exists $\mu_1, ..., \mu_{k+1}$ with sum 1 such that $0 = \sum_i \mu_i \lambda_i$. We then see that all λ_i cannot be strictly negative at the same time.

We now show that if $J \neq [\![1, k+1]\!]$, the intersection is not empty. Condition (C3) gives us linear independance of the family $(\lambda_j)_{j \in J}$. The linear system $\lambda_j(y) = -1, j \in J$ then has a solution, where the unknown is y. □

Because every C_i is convex, their intersections are contractible. Applying Leray's nerve theorem (cite), $\lambda^{-1}(-1)$ has the homotopy type of the simplical complex given by the nerve. By virtue of what we discussed, this complex is exactly $K(S_1^c, ..., S_l^c)$. The homotopy type of $f^{-1}(-1)$ is then given by

$$f^{-1}(-1) \simeq K(S_1^c, ..., S_l^c) * \mathbb{S}^\mu$$

By recalling that a join with a sphere is a suspension, we get the following result in homology.

Theorem 5. *For $i \geq 0$, we have*

$$H_{i+\mu}(M_1, M_{-1}) = \tilde{H}_{i-1}(K(S_1^c, ..., S_k^c))$$

3 First properties of the euclidean distance

In this section, we will work in the euclidean space $\mathbb{R}^m$. Some of what we'll do here can be generalized to a more general Riemannian manifold. All throughout, $X \subset \mathbb{R}^m$ will be a smooth manifold unless specified otherwise, $Y \subset \mathbb{R}^m$ a closed set.

The main function we will study is the euclidean distance from Y restricted to X, i.e

$$\forall x \in X, \mathrm{dist}_Y(x) = \min_{y \in Y} |x - y|$$

where $|\cdot|$ is the euclidean norm. This function is locally Lipschitz. In what will follow, we identify $(\mathbb{R}^m)^*$ with $\mathbb{R}^m$ via the euclidean structure. $\partial_x(\mathrm{dist}_{Y|X})$ will sometimes be identified with a subset of T_xX, where $x \in X$.

Our main objectives will be to understand the subdifferential of this function and to find a setting where Morse theory of continuous selections can be applied.

3.1 The subdifferential of the distance function

Our goal in this section will be to compute explicitly the subdifferential of $\mathrm{dist}_{Y|X}$, which gives rise to a very nice and intuitive geometric picture. We first require some results on "infinite continuous selections", which is indeed how dist_Y is defined.

Let $\varphi : \mathbb{R}^m \times Z \to \mathbb{R}$ be a continuous function, where Z is a compact space. We can define a family $(\varphi_z)_{z \in Z}$ of continuous functions, where $\varphi_z = \varphi(\cdot, z)$. Similar to the finite case we can define a "compact continuous selection" as a continuous f such that for all $x \in \mathbb{R}^m$

$$f(x) \in \{\varphi_z(x) \mid z \in Z\}$$

Similarly to the finite case, we define the set of active indices at x, $Z(x)$ as the set of all z that take the value of f at x. The following result is a useful generalization of a property of finite continuous selections.

Lemma 8. *The function $x \mapsto Z(x)$ is upper semi-continuous.*

Proof. Let $x \in \mathbb{R}^m$ and let V be an open neighbourhood of $Z(x)$. Consider for all $n \in \mathbb{N}$ the closed ball B_n of center x and radius 2^{-n}. Set

$$A_n = \{z \in Z \mid \exists y \in B_n, \varphi_z(y) = f(y)\}$$

A_n is closed : indeed if we write $g = \varphi - f$, we can write A_n as $\mathrm{proj}_Z(g^{-1}(0) \cap (B_n \times Z))$, and $g^{-1}(0) \cap (B_n \times Z)$ is compact so we get what we want.

Note that the intersection $\bigcap_n (A_n \backslash V)$ is empty : if there was a point z in such intersection, there would be a sequence $x_k \to x$ along which $\varphi_z(x_k) = f(x_k)$, so $z \in Z(x) \subset V$. By compactness, and because (A_n) is decreasing, there is some n such that $A_n \subset V$. Let $y \in B_n$ for such n. We have

$$Z(y) \subset A_n \subset V$$

so we get the desired result. □

The following theorem will allow us to compute explicitly subdifferentials in a particular case.

Theorem 6 (Danskin's [3])**.** *Assume that for all z, φ_z is differentiable, $z \mapsto d_x\varphi_z$ is continuous and $f = \max_{z \in Z} \varphi_z$.*

1. *f is convex and for all $x, v \in \mathbb{R}^m$*
$$D_x f(v) = \max_{z \in Z(x)} d_x\varphi_z(v)$$

2. *f is locally Lipschitz and we have*

$$\partial_x f = \operatorname{conv}\{d_x\varphi_z(v) \mid z \in Z(x)\}$$

Proof. We'll need 1. for 2.

1. The convexity of f is a classical result from convex analysis. Notice that for $t > 0$

$$\frac{f(x+tv) - f(x)}{t} \geq \frac{\varphi_z(x+tv) - \varphi_z(x)}{t}$$

for $z \in Z(x)$, so we get $D_x f(v) \geq d_x\varphi_z(v)$. For the other inequality, consider a positive sequence $t_n \to 0$. For all n, select $z_n \in Z(x + t_n v)$, by compactness assume it converges to $z^* \in Z$. By maximality of f, we have $\varphi_{z^*}(x) \geq \varphi_z(x), z \in Z$, so $z^* \in Z(x)$. By convexity of f and φ_{z_n}, we get

$$\begin{aligned} D_x f(v) &\leq \frac{f(x+t_n v) - f(x)}{t_n} \\ &= \frac{\varphi_{z_n}(x+t_n v) - \varphi_z^*(x)}{t_n} \\ &\leq \frac{\varphi_{z_n}(x+t_n v) - \varphi_{z_n}((x+t_n v) - t_n v)}{t_n} \\ &\leq d_x\varphi_{z_n}(v) \\ &\to d_x\varphi_{z^*}(v) \end{aligned}$$

so $D_x f(v)$ is indeed the maximum.

2. All φ_z are locally Lipschitz. By compactness of Z, we can find a Lipschitz constant for f.

We now show the equality involving ∂f. For that we use the results of the discussion in section 1.1.2. For the right to left inclusion, it suffices to observe that in virtue of 1., for $z \in Z(x)$

$$\forall v \in \mathbb{R}^m, f(x+v) \geq f(x) + d_x\varphi_z(v)$$

Assume that there is some $l \in \partial_x f$ not in the convex hull of $d_x\varphi_z$, $z \in Z(x)$. Because this convex hull is compact, we choose a strictly separating hyperplane : we choose v and c such that

$$\langle l, v\rangle > c > \langle d_x\varphi_z, v\rangle, z \in Z(x)$$

This contradicts the fact that $D_x f(v) = \max_{h\in\partial_x f}\langle h, v\rangle$, because $D_x f(v) = \max_{z\in Z(x)}\langle d_x\varphi_z, v\rangle$

□

We are now ready to compute the subdifferential of $\operatorname{dist}_{Y|X}$.

Theorem 7. *Let $x \in X\backslash Y$. We have*

$$\partial_x(\operatorname{dist}_{Y|X}) = \operatorname{proj}_{T_xX}\left(\operatorname{conv}\left\{\frac{x-y}{|x-y|} \mid y \in Y(x)\right\}\right)$$

where $\operatorname{proj}_{T_xX}$ *is the orthogonal projection onto* T_xX.

Proof. We first compute the subdifferential in the case where $X = \mathbb{R}^m$, and we deduce the general case. Let $\eta_Y = \frac{1}{2}\operatorname{dist}_Y^2$. We write for $x \in \mathbb{R}^m\backslash Y$

$$\eta_Y(x) = \frac{1}{2}|x|^2 - \left(\max_{y\in Y}\langle x, y\rangle - \frac{1}{2}|y|^2\right)$$

The function $\varphi_y(x) = \langle x, y\rangle - \frac{1}{2}|y|^2$ is convex. If we denote by $h(x)$ the term in parenthesis, theorem 6 yields

$$\partial_x h = \operatorname{conv}\{\nabla_x\varphi_y \mid y \in Y(x)\} = \operatorname{conv}(Y(x))$$

We can apply the theorem because in a small neighbourhood of x, the active indices stay confined to a compact set.
By using that $\eta_Y = \frac{1}{2}|\cdot| - h$, we get

$$\partial_x \eta_Y = \{x - y \mid y \in Y(x)\}$$

If we now write $\mathrm{dist}_Y = \sqrt{2\eta_Y}$, and use the subdifferential chain rule applied to the diffeomorphism $\sqrt{\cdot}$ (because $\eta_Y(x) \neq 0$, $x \notin Y$) we get

$$\begin{aligned}\partial_x \mathrm{dist}_Y &= \frac{1}{\sqrt{2\eta_Y(x)}} \mathrm{conv}\{x - y \mid y \in Y(x)\} \\ &= \mathrm{conv}\left\{\frac{x-y}{|x-y|} \mid y \in Y(x)\right\}\end{aligned}$$

because $\eta_Y(x) = \frac{1}{2}|y - x|^2$ for all $y \in Y(x)$.

Denote $g = \mathrm{dist}_Y$, and $f = \mathrm{dist}_{Y|X}$. We will show that $\partial_x f = \mathrm{proj}_{T_xX}(\partial_x g)$.

We start from left to right. We want to show that $x \mapsto \mathrm{proj}_{T_xX}(\partial_x \mathrm{dist}_Y)$ is upper-semi continuous (where the set is seen in TX), and contains $\nabla_x f$ whenever defined. The first part is easy by semi-continuity of the original function. For the second, we assume $\nabla_x f$ is defined for some $x \in X$. By the computations made earlier and Danskin's theorem, we write

$$D_x g(v) = \min_{y \in Y(x)} \langle \nabla_x g_y, v\rangle$$

where $g_y(x) = |x - y|$. Note that because $\nabla_x f$ exists, this is linear if restricted to T_xX. By using linearity, we find that there is $y \in Y(x)$ such that for all $v \in T_xX$

$$D_x g(v) = \langle \nabla_x g_y, v\rangle = \langle \nabla_x f, v\rangle$$

Hence $\nabla_x g_y$ projects onto $\nabla_x f$, so we have what we want. The inclusion then follows from minimality of subdifferentials.

For the inclusion from right to left, we consider $x_k \to x \in X$ a sequence such that $\nabla_{x_k} g$ is well defined and converges, and we will show that it is a subgradient of $\mathrm{dist}_{Y|X}$ up to projection. We consider up to restricting to a neighbourhood of x, that $X = \mathbb{R}^n$ where $n = \dim(X)$. Let $(V_k)_k$ be a decrasing sequence of open neighbourhoods of $\partial_x f$ converging to it. By semi-continuity of ∂g and up to extracting, $\nabla_{x_k} g \in V_k \times \mathbb{R}^{m-n}$, So the projection of $\nabla_{x_k} g$ converges to a point in $\partial_x f$, we got equality. □

3.2 Special points of the distance function

We now consider $X \subset \mathbb{R}^m$ to be some non specific submanifold. Following [2], we define

Definition 5 (Medial axis)**.** *We define M_Y, the medial axis of Y as*

$$M_Y = \{x \in \mathbb{R}^m \mid |Y(x)| \geq 2\}$$

Some other special points of the distance function include the cut-locus of Y, that we'll denote $\mathrm{CLoc}(Y)$ and the central set of Y that we denote $C(Y)$, introduced respectively by Thom [8] and Milman [4].

Definition 6 (Central set)**.** *An open ball $B \subset \mathbb{R}^m \backslash Y$ will be called maximal if there is no open ball in $\mathbb{R}^m \backslash Y$ strictly containing B. The central set $C(Y)$ is then the set of the centers of all maximal balls.*

If Y is smooth we can also define the following set

Definition 7 (Cut-locus)**.** *The cut-locus of Y is defined as follows*

$$\mathrm{CLoc}(Y) = M_Y \cup \{x \in \mathbb{R}^n \mid y \in Y \mapsto |x - y|^2 \textit{ has a degenerate global minimum}\}$$

If we assume that Y is a submanifold of $\mathbb{R}^m$, we can link these seemingly differents concepts : in this case, $\overline{M_Y} = \mathrm{CLoc}(Y) = C(Y)$. $\overline{M_Y} = C(Y)$ is proved for Y defined as the border of a bounded domain in [4], but the arguments can be adapted to a more general case. The same kind of arguments involving curvatures give us that $\mathrm{CLoc}(Y) \subset C(Y)$. Lastly by considering a sequence with limit in $\overline{M_Y} \backslash M_Y$, one can show that $\overline{M_Y} \subset \mathrm{CLoc}(Y)$.

Using similar computations as in theorem 7, we can show, because dist_Y and its square have the same differentiability points out of Y, that outside of $Y \cup M_Y$ by Danskin's theorem, dist_Y is differentiable. In fact we have the following stronger statement.

Proposition 4. *The function* dist_Y *is* C^1 *on* $\mathbb{R}^m \backslash (Y \cup \overline{M_Y})$.

Proof. Let $x \in \mathbb{R}^m \backslash (Y \cup \overline{M_Y})$. Let U be a neighbourhood of x contained in $Y \cup \overline{M_Y}$. On this neighbourhood, we can then define a function $\eta : U \to Y$ that to each $z \in U$ associates the unique $y \in Y(z)$. It follows from semi-continuity of the active index that η is continuous. Denote by $g : (a, b) \mapsto |a - b|$ and $g_a = g(a, \cdot)$. dist_Y is differentiable on U. Using the first part of Danskin's theorem and similar computations to theorem 7, we get

$$d_z \,\mathrm{dist}_Y = d_z g_{\eta(z)}$$

From this, we deduce that $d\,\mathrm{dist}_Y$ is continuous on U. □

3.3 Reducing to local continuous selections

In this section, $Y \subset \mathbb{R}^m$ will be a smooth submanifold, as X. Our goal will be to show that under certain assumptions, the distance function is locally a continuous selection of finitely many C^2 functions, and so enters the Morse theory framework developped in section 2.

We'll denote by φ the Riemannian exponential map restricted to NY, i.e for $(y, v) \in NY$, $\varphi(y, v) = y + v$. The following proposition from [2] uses this map to find points where dist_Y is locally a continuous selection.

Proposition 5. *Assume* Y *is smooth and let* $x \in \mathbb{R}^m$ *be a regular value of* φ. *Then* $Y(x) = \{y_1, ..., y_n\}$ *is finite. Moreover there exists*

- V *a neighbourhood of* x
- *For all* i, W_i *a neighbourhood of* y_i *such that* $\overline{W_i}$ *are disjoint*

that verify

$$\forall z \in V, \mathrm{dist}_Y(z) = \min_i \mathrm{dist}_{Y \cap \overline{W_i}}(z)$$

and all $\mathrm{dist}_{Y \cap \overline{W_i}}$ *are smooth on* V.

Proof. We first prove that $Y(x)$ is finite. This set is compact. If we write $\pi : NY \to Y$ the natural projection, we get $Y(x) \subset \pi(\varphi^{-1}(x))$. But note that x being a regular value, $\varphi^{-1}(x)$ is of codimension m in NY, hence it is a set of disjoint points. The same applies after applying π, so $Y(x)$ is finite.

We now construct the neighbourhoods : denote $Y(x) = \{y_1, ..., y_n\}$. For all i, because x is regular, we choose a neighbourhood U_i of $x - y_i$ in $\mathbb{R}^{m-\dim(Y)} \cong N_{y_i}Y$, W_i a neighbourhood of y_i in $\mathbb{R}^m$, and V_i a neighbourhood of x in $\mathbb{R}^m$ such that, up to trivialization, $(W_i \cap Y) \times U_i \subset NY$, and $\varphi : (W_i \cap Y) \times U_i \to V_i$ is a diffeomorphism.

Now note that for all z in some neighbourhood $V \subset \bigcap_i V_i$ of x, $z - y$ is close to $x - y_i$ for all $y \in W_i \cap Y$. So if it happens that $z - y \in N_y Y$, then $z - y \in U_i$. From that observation, and the induced diffeomorphism, we deduce that there is only one such y. From that we get

$$\forall z \in V, \mathrm{dist}_{Y \cap \overline{W_i}}(z) = |\pi(\varphi_i^{-1}(z)) - z|$$

where $\varphi_i := \varphi_{|(W_i \cap Y) \times U_i}$. So the distance is indeed a smooth function on V.

Lastly, by semi-continuity of $x \mapsto Y(x)$, we can assume that V is small enough such that for $z \in V$, $Y(z) \subset \bigcup_i W_i$. Hence

$$\mathrm{dist}_Y(z) = \min_i \mathrm{dist}_{Y \cap \overline{W_i}}(z)$$

□

A natural question following this result would be to consider how restrictive is the condition of being a regular value of φ, i.e what are these points, is it likely that a critical point of $\mathrm{dist}_{Y|X}$ is not a regular value and what can we say in this case.

We can first observe that critical values of φ are points in $\mathbb{R}^m$ verifying a degeneracy condition similar to the one defining the cut-locus $\mathrm{CLoc}(Y)$: x is a critical value if there is $y \in Y(x)$ that is a degenerate minimum of the function $y' \in Y \mapsto |x - y'|$. These points do not match $\mathrm{CLoc}(Y)$ though, as a regular value x of φ can be such that $Y(x)$ is not a singleton.

If we try to generalize the statement by dropping the regular value condition, we can still show using the semi-continuity of the active index and compactness that there is finitely many neighbourhoods W_i such that for z close enough to x, $\mathrm{dist}_Y(z)$ can still be obtained by minimizing over $\mathrm{dist}_{Y \cap \overline{W_i}}(z)$.
We do not get smoothness of these distances near x for arbitrary Y, a counterexample is given by $Y = \mathbb{S}^{m-1}$ and $x = 0$: if for some choice of W_i, the distances were smooth, then $\partial_0(\mathrm{dist}_{\mathbb{S}^{m-1} \cap \overline{W_i}})$ would consist of a single point, which is not the case by theorem 7.

On the "likelyhood" side, it is shown that when Y is algebraic, then in the generic case, critical points of the distance do not intersect critical values of φ [2].

4 Algebraic point of view on the euclidean distance

In this section, we still place ourselves in the euclidean space $\mathbb{R}^m$. We will consider $X, Y \subset \mathbb{R}^m$ algebraic manifolds or varieties. We still study the distance from Y restricted to X, $\text{dist}_{Y|X}$. Our goal will be to study the critical points of this function from an algebraic point of view.

4.1 The critical point equation

We first introduce some notation : Y is defined by the zero set of polynomials $q_1, ..., q_n$, while X is defined by the zero set of $p_1, ..., p_r$. We assume that these are manifolds, which can be done without much trouble for what will come by restricting to the smooth locus of X and Y.

From theorem 7 proved in the last section, we now that a point $x \in X$ is critical for the distance function iff there exists a number $k \geq 1$, $y_1, ..., y_k \in Y$ and $\lambda_1, ..., \lambda_k \in \mathbb{R}$ such that :

1. The y_i are all distinct
2. $|x - y_i| = \text{dist}_Y(x)$ for all i
3. $\sum_i \lambda_i \frac{x-y_i}{|x-y_i|} \in N_x X$
4. For all i, $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$.

We want to transform these into algebraic equations, such that the solutions of these new equations include true critical points of the distance function. The following equations with unknowns $x, y_1, ..., y_k$ accomplish that

$$\begin{array}{ll} 1. & \text{The } y_i \text{ are all distinct} \\ 2. & \forall i, j, |x - y_i|^2 = |x - y_j|^2 \text{ and } x - y_i \in N_{y_i} Y \\ 3. & \text{The family } (x - y_i)_i \text{ is linearly dependent in } \mathbb{K}^m / N_x X \end{array} \tag{KCP}$$

where $\mathbb{K} = \mathbb{R}$ for now.
The first condition is an open one. The second one is algebraic because $N_{y_i} Y$ is spanned by $\nabla_{y_i} q_1, ..., \nabla_{y_i} q_n$, so $x - y_i \in N_{y_i} Y$ is equivalent to asking that the family $\{x - y_i, \nabla_{y_i} q_1, ..., \nabla_{y_i} q_n\}$ is linearly dependent. For similar reasons we can show that the last condition is also agebraic.
It is also easy to check that true critical points provide a solution to these equations, as for y_i to minimize the distance from x to Y, one needs $x - y_i \in N_{y_i} Y$.

The next step to better understand these polynomials equations is to view them as complex. For that, we consider the complexification of X and Y : $X_{\mathbb{C}}$ is the submanifold of $\mathbb{C}^m$ defined as the zero set of the complex polynomials $p_1, ..., p_r$, and similarly for $Y_{\mathbb{C}}$. We also need to define what it means for some complex vector to be in the normal bundle of $X_{\mathbb{C}}$ (or $Y_{\mathbb{C}}$) : the normal bundle over $X_{\mathbb{C}}$ is defined as the vector bundle whose fiber at all $x \in X_{\mathbb{C}}$ are

$$N_x X_{\mathbb{C}} = \{v \in \mathbb{C}^m \mid \forall w \in T_x X_{\mathbb{C}}, \sum_{i=1}^{m} v_i w_i = 0\}$$

Note that the product involved is not an inner product on $\mathbb{C}^m$ but has the advantage of being algebraic. It can be checked that this vector space is spanned by the gradients of $p_1, ..., p_r$, so the observations made in the real case still hold. With these definitions, we can then replace $\mathbb{K}$ with $\mathbb{C}$ in (KCP), and for what follows we will exclusively study this complex version. We will then shorten $X_{\mathbb{C}}, Y_{\mathbb{C}}$ by X, Y, considering them as already complexified.

In what follows, we will study particular cases of these equations and connect them to already existing notions on algebraic varietes/manifolds (?). The case where $Y = \{y\}$ will lead to the notion of *euclidiean distance degree*, and when $k = 2$ and $X = \mathbb{C}^m$ we will get the notion of *bottleneck degree*.

4.2 Euclidean distance degree

4.2.1 Definition and examples

As stated earlier, we consider the case where $Y = \{y\}$ and X is some algebraic submanifold of $\mathbb{C}^m$. (KCP) then simplifies to

$$x - y \in N_x X$$

where the unknown is x. This is equivalent to asking that $y \in x + N_x X$. We set $\hat{N}_x X := x + N_x X$ and define the following bundle on X

$$\hat{N}X = \{(x, v) \mid x \in X, v \in \hat{N}_x X\} \subset X \times \mathbb{C}^m$$

This bundle has two natural projections $\pi_1 : \hat{N}X \to X$ and $\pi_2 : \hat{N}X \to \mathbb{C}^m$. The set of solutions to (KCP) is then $\pi_1(\pi_2^{-1}(y))$. Using the fact that $\hat{N}X$ is of dimension m, one can show the following property.

Proposition 6. *For generic $y \in \mathbb{C}^m$, the set $\pi_1(\pi_2^{-1}(y))$ is finite and it has a constant number of elements.*

The *euclidean distance degree* (EDD) of X is then defined as this generic number of solutions to the critical point equation. It then gives an upper bound on the number of real solutions to (KCP).

Consider the example where $X \subset \mathbb{C}^2$ is an ellipse. It is then defined by a degree 2 polynomial and we will show later that the euclidean distance degree of X is 4. We see that for generic points y in the bounded region of the plane $\mathbb{R}^2$ created by X, we get the maximum amount, 4 real critical points, and only 2 when y is in the unbounded one.
Not all submanifolds X can have as many real critical point as the euclidean distance degree, as can be seen with the parabola, of euclidean distance degree 4 : for all points y, we have at most 3 real critical points.
(draw something)

4.2.2 Chow ring and useful classes

We now present some tools that will help us compute the euclidean distance degree of a reduced, irreducible complex projective variety $X \subset \mathbb{P}^m$ (correct if this is wrong). Let it be defined as the projective zero set of some polynomials, $p_1, ..., p_{m-n}$. X is then of dimension n.

Definition 8. *The group of k-cycles of X, where $k \leq n$ is the free abelian group generated by the $Y \subset X$ that are closed, k-dimensional subvarieties denoted $Z_k(X)$.*

Consider $W \subset X$ a $k+1$-dimensional subvariety of X and f a rationnal function over W. One can, for $Z \subset W$ a k-dimensional variety, define the order of vanishing of f along Z, denoted $\mathrm{ord}_Z(f)$ (cite) : intuitively it corresponds to the exponent $c \in \mathbb{Z}$ such that near a generic point x in Z, f behaves as some non zero factor times a degree c monomial. One then defines the divisor of f as

$$\mathrm{div}(f) = \sum_{\substack{Z \subset W \\ \dim(Z)=k}} \mathrm{ord}_Z(f) Z$$

If we denote $R_k(X)$ the subgroup of $Z_k(X)$ generated by the divisors of rationnal functions over some $k+1$ dimensional $W \subset X$, we get the following definition of the Chow group.

Definition 9. *The Chow group of X is defined as*

$$A(X) = \bigoplus_k A_k(X)$$

where for all k, $A_k(X) = Z_k(X)/R_k(X)$

As an example, if $X = \mathbb{P}^2$ then, $[\{x^2 + y^2 - z^2 = 0\}] = 2[\{x = 0\}]$ in $A(\mathbb{P}^2)$ because the following function, $f(x, y, z) = \frac{x^2+y^2-z^2}{x^2}$ is a rationnal function over a subvariety of $\mathbb{P}^2$ such that its divisor gives exactly the difference of these two terms.

We now make this group a ring via the following theorem (cite)

Theorem 8.

4.2.3 Computing the EDD

4.3 Bottleneck degree

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